

4. An incomplete diagram of the electromagnetic (em) spectrum is shown below.

Region	Radio waves	Microwaves	Infra-red	Visible
Wavelength (m)				

(a) The wavelength of a wave in each of the regions shown is given below.
Complete the table by using the wavelength values below. [2]

$2 \times 10^{-2} \text{ m}$ $4 \times 10^2 \text{ m}$ $5 \times 10^{-7} \text{ m}$ $3 \times 10^{-5} \text{ m}$

(b) The ionising regions of the em spectrum are not included on the diagram.
(i) Name the missing regions. [1]

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(ii) State what is meant by the term ionising radiation. [1]

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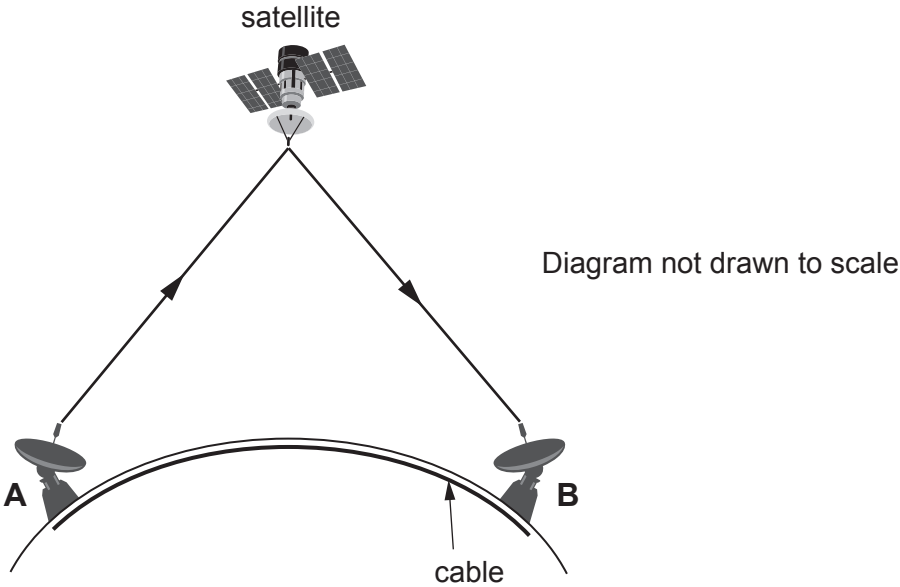
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(c) All waves in the em spectrum are transverse.
State **one** other property they have in common. [1]

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- (d) Electromagnetic radiation is used in communications.
A and B are two points on the Earth's surface.
To send a signal from A to B, it can either be sent as a signal in a cable or as a signal via a satellite in geostationary orbit.



Information about both methods is given in the table below.

Speed of signal to satellite	$3 \times 10^8 \text{ m/s}$
Speed of signal in cable	$2 \times 10^8 \text{ m/s}$
Distance from A to B through the cable	$6 \times 10^6 \text{ m}$
Distance from A to B via the satellite	$7.2 \times 10^7 \text{ m}$

Owain states that it is better to send the signal by satellite as it travels faster so it will arrive in a shorter time.
Use data from the table to explain whether you agree.

[3]

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